

PAPER

Performance Analysis of CPDMA-Based Satellite-Terrestrial Mobile Networks with Power Splitting Enabled Incremental Relaying

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SUMMARY A cooperative pattern division multiple access with power splitting enabled incremental relaying (PSEIR-CPDMA) is proposed for downlink satellite-terrestrial mobile networks. The strongest user decides whether to operate cooperative communication according to the feedback of the weak users. When in CPDMA mode, the strongest user as a power splitting enabled incremental relay node uses the harvested energy to relay signals. Under delay-limited transmission condition, the outage probability and throughput of the proposed system are investigated in depth, and the theoretical analysis is verified by simulation for performance evaluation. The results demonstrate that the PSEIR-CPDMA scheme can achieve better performance than the conventional CPDMA with PS (PS-CPDMA).

key words: cooperative pattern division multiple access (CPDMA), satellite-terrestrial mobile networks (STMN), energy harvesting (EH), power splitting (PS), incremental relaying (IR)

1. Introduction

Currently, International Telecommunication Union (ITU) has identified satellite communications as an important part of 6G integrated ManyNets. The overall vision of 6G is integrating satellite mobile communication, artificial intelligence and big data to create an “air-sky-sea-land” integrated network based on the further expansion and upgrading of 5G network. The application scenarios of 6G network will be upgraded and expanded from the three application scenarios of 5G to the five application scenarios. Therefore, in order to realize efficient, fast and intelligent access, it is of great significance to study and explore satellite communication access technology in 6G network.

As an emerging technology in spectrum utilization of 5G cellular communication, non-orthogonal multiple access (NOMA) technology is applied in satellite communication to solve the problem of resource shortage and spectrum utilization conflict, which has become one of the hottest spots in the theoretical research of satellite communication technology [1], [2]. To further expand coverage and overcome the decline of wireless transmission, cooperative NOMA technology (CNOMA) has attracted the attention of some

scholars [3], [4]. CNOMA is introduced into a two-layer geostationary Earth orbit/low Earth orbit satellite network in a frequency coexistence scenario and the performance of the considered network is investigated [5].

Pattern division multiple access (PDMA), which as a multi-carrier NOMA, can simultaneously utilize resources in the frequency/time/spatial domain. Significantly, cooperative pattern division multiple access (CPDMA) has been shown to have many performance advantages [6], [7], but a separate DF relay node will bring additional energy consumption and economic cost.

In conventional CNOMA, the strong user often needs to provide additional energy consumption for relaying, which will burden the strong user. Therefore, considering the energy consumption of the strong user as a relay, simultaneous wireless information and power transfer (SWIPT) is applied to CNOMA for the first time [8]. In a two-user CNOMA system with SWIPT, a best-near best-far (BNBF) user selection scheme is proposed and the outage performance is evaluated [9]. In addition, for multi-relay cooperative cognitive radio NOMA (CR-NOMA) with SWIPT, a two-stage relay selection strategy is introduced, and the impact on outage performance is analyzed through parameters such as power splitting factor [10]. However, the fairness of users is not considered in these schemes.

In order to further improve the fairness among users and save the energy consumption of strong users, an incremental relay (IR) protocol is proposed, in which 1-bit feedback message is used to decide whether to adopt direct or cooperative transmission [11]. In addition, the optimal relay node is selected using harmonic mean function, and the optimal power is allocated between the ground station and other mobile users including cooperative relay node [12]. And an energy harvesting (EH) IR-CNOMA system with an optimal PS factor is constructed to ensure maximum throughput [13]. If the above solutions can be integrated and optimized, the system performance will be further improved.

In this paper, we propose PSEIR-CPDMA scheme and apply it to satellite mobile network. In the scheme, the strong user acts as an half-duplex (HD) IR to choose either direct PDMA (DPDMA) or CPDMA mode according to the weak users' feedback signals, and IR operates EH and PS protocol to provide energy for decode-and-forward (DF) relaying, so as to reduce the burden of the strong user and improve the energy efficiency. The primary contribu-

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tions of this paper are summarized as follows: 1) First, IR-CPDMA combined with EH and PS protocols is applied to STMN, whose satellite-terrestrial link and terrestrial link obey Shadowed-Rician fading distribution and Nakagami-m distribution respectively. 2) The expressions of outage probability for each user and system throughput are derived in delay-limited mode. 3) Finally, compared with conventional CPDMA with PS (PS-CPDMA), the proposed scheme can greatly improve outage performance, system throughput and energy efficiency.

2. System Model

In the single satellite multi beam scenario, a CPDMA based STMN is considered as shown in Fig. 1. The satellite uses array fed reflector antenna to form multi beam coverage. Within the beam, the ground users are divided into multiple clusters according to the feedback channel state information (CSI). Different clusters multiplex orthogonal communication resources, while K users within a cluster can occupy non-orthogonal physical resources. For example, the satellite uses PDMA to communicate with K users in the cluster by pattern matrix $G_{N \times K}^{PDMA}$. $G_{N \times K}^{PDMA}$ represents the mapping from transmitted data of K users to N available resource elements (REs) with $K > N$.

Given the proposed communication strategy, the users within a cluster need to be sorted based on the line-of-sight link gains. Without loss of generality, assume the K users are sorted as $\|h_K^{su}\|^2 > \|h_{K-1}^{su}\|^2 > \dots > \|h_1^{su}\|^2$, where $h_k^{su} = (h_{n,k}^{su})_{N \times 1}$ is a channel fading vector from satellite (S) to User i (U_i) on N REs. $\|\cdot\|$ represents the Frobenius norm of the matrix. Supposing each node is equipped with a single antenna, the antenna gain of each mobile user is 1.

2.1 Channel Model

From the system model, it can be seen that there are two different types of channel fading. The satellite-terrestrial link budget for $S \rightarrow U_i$ ($i = 1, 2, 3$) on the n^{th} RE is modeled as

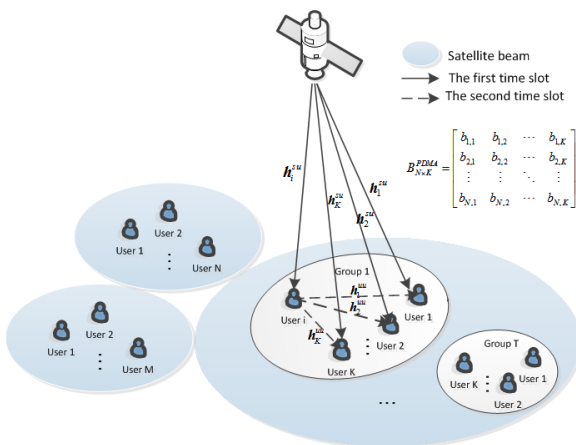


Fig. 1 PSEIR-CPDMA downlink STMN.

$$H_i = F_i G_s B(\varphi_i) |h_{n,i}^{su}|^2, \quad (1)$$

where $F_i = \left(\frac{\lambda}{4\pi d_i}\right)^2$ with λ and d_i being operating wavelength and the distance from S to U_i . The beam gain $B(\varphi_i) = (0.5J_1(u_i)/u_i + 36J_3(u_i)/(u_i)^3)^2$ depends on the satellite beam pattern and the user's location [14], therein $J_n(\cdot)$ denotes the first kind Bessel function of the n^{th} order. $u_i = 2.07123 \sin \varphi_i / \sin \varphi_{i3dB}$ with φ_i and φ_{i3dB} being the included angle and the 3-dB angle between U_i and the beam center line. G_s denotes the satellite antenna gain.

The $S \rightarrow U_i$ link adopts a widely-used Shadowed-Rician fading mode for S-band [15], the probability density function (PDF) of $|h_{n,i}^{su}|^2$ ($i = 1, 2, 3$) is given as

$$f_{|h_{n,i}^{su}|^2}(x) = \rho_i e^{-\beta_i x} {}_1F_1(\tau_i; 1; \delta_i x), \quad (2)$$

where $\rho_i = (2b_i\tau_i)^{\tau_i} / 2b_i(2b_i\tau_i + \Omega_i)^{\tau_i}$, $\delta_i = 0.5\Omega_i/b_i/(2b_i\tau_i + \Omega_i)$, $\beta_i = 0.5/b_i$, with $2b_i$ and Ω_i being the average power values of the multi-path component and the light-of-sight component respectively. τ_i ($\tau_i > 0$) is Nakagami-m fading parameter. ${}_1F_1(a; b; c)$ denotes the confluent hypergeometric function.

As according to [16], the cumulative distribution function (CDF) of $|h_{n,i}^{su}|^2$ ($i = 1, 2, 3$) to be used later is derived as

$$F_{|h_{n,i}^{su}|^2}(u) = \rho_i \sum_{k=0}^{\infty} \frac{(\tau_i)_k \delta_i^k}{(k!)^2 \beta_i^{k+1}} \gamma(k+1, \beta_i u). \quad (3)$$

Assuming τ_i is an integer as in [17], Eq. (3) can also be expressed as

$$F_{|h_{n,i}^{su}|^2}(u) = \rho_i \sum_{k=0}^{\tau_i-1} \Pi_{i,k} \gamma(k+1, (\beta_i - \delta_i)u), \quad (4)$$

where $\Pi_{i,k} = \frac{(-\delta_i)^k (1 - \tau_i)_k}{(k!)^2 (\beta_i - \delta_i)^{k+1}}$, $\gamma(a, x) = \int_0^x e^{-t} t^{a-1} dt$ is called low-order nonholonomic Gamma function. $(x)_k = \Gamma(x+k)/\Gamma(x)$ is Pochhammer symbol with $\Gamma(x)$ being Gamma function.

The terrestrial channel power gain factor for $U_3 \rightarrow U_j$ ($j = 1, 2$) is given as

$$X_j = E_j |h_j^{uu}|^2, \quad (5)$$

where $E_j = 10^{-\frac{L_j}{10}}$, and L_j [dB] = $42 + 22 \lg l_j$ [Km] + $20 \lg(f$ [GHz]/5) [18] represents wireless channel free space loss (FSL) with l_j and f being the distance and operating frequency from U_3 to U_j .

The terrestrial link $|h_j^{uu}|^2$ ($j = 1, 2$) obeys Nakagami-m distribution, whose PDF is expressed as

$$f_{|h_j^{uu}|^2}(y) = \frac{\varepsilon_j^{m_j} y^{m_j-1}}{\Gamma(m_j)} e^{-\varepsilon_j y}, \quad (6)$$

where $\varepsilon_j = \frac{m_j}{\nu_j}$ with m_j and ν_j being fading parameter and

the average power of the link.

2.2 Channel Model

Here the K users multiplex on N physical resource elements (REs) as a PDMA group with a binary indicative pattern matrix $G_{N \times K}^{PDMA} = [\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_K]$ [21], the coding pattern $\mathbf{g}_i$ denotes the mapping of U_i onto REs. The satellite sends a superposed signal $\mathbf{x}_s = G_{N \times K}^{PDMA} [x_1 \ x_2 \ \dots \ x_K]^T = \sum_{i=1}^K \sqrt{\alpha_i P} \mathbf{g}_i x_i$ to each user, in which α_i ($\sum_{i=1}^K \alpha_i = 1$) is a power scaling factor for U_i , and the downlink sending power is P at S. It is noticed that a dynamic transmitting power allocation (DTPA) scheme is adopted, which can allocate higher power to weaker users to improve throughput and keep fairness. The value of power scaling factor is adjusted in time according to the user's channel conditions as in [19]. The parameter x_i ($E(|x_i|^2) = 1$) is the modulated symbol for U_i .

At the beginning of transmission, the satellite broadcasts a pilot signal to each user. After receiving the signal, the user calculates the received signal to noise ratio (SNR), then it returns an ACK or NACK to the satellite and U_k according to the relative size of SNR and the given threshold. Here assuming that U_k with optimal channel quality always returns an ACK, so it can be used as an incremental HD relay to decode and forward signals for the other users. When all users return ACKs, all direct links are valid and the network operates in direct PDMA (DPDMA) mode.

If any user returns a NACK, U_k enables EH and PS protocol to forward data as a relay, and the network works in cooperative PDMA mode. A PS receiver is configured at the U_k receiving antenna, so the received signal is divided into two independent branches: $1 - \rho$ proportion is used for signal reception, and ρ proportion is used for energy harvesting. $0 \leq \rho \leq 1$ represents the power splitting factor, and the collected energy is transferred to the battery as temporary storage used to forward information, saving energy consumption and improving the energy limitation problem of cooperative relay nodes. But other users who return ACKs still use direct transmission.

With DPDMA, the satellite sends a PDMA superposed signal to each user. So the received signal at User k is

$$\begin{aligned} \mathbf{y}_k^{su} &= [\sqrt{q_{1,k}} h_{1,k}^{su} \ \sqrt{q_{2,k}} h_{2,k}^{su} \ \dots \ \sqrt{q_{N,k}} h_{N,k}^{su}]^T \odot \mathbf{x}_s + \mathbf{n}_k^{su} \\ &= \sum_{i=1}^K \sqrt{\alpha_i P} x_i \bar{\mathbf{h}}_i^{su} + \mathbf{n}_k^{su} \end{aligned} \quad (7)$$

where $q_{n,k}$ denotes the channel attenuation coefficient on the n^{th} RE for $S \rightarrow U_k$. For simplicity, assuming that the channels among the REs are completely related, $h_{n,k}^{su} = h_k^{su}$ and

$$q_{n,k} = q_k = F_k G_S B(\varphi_k). \text{ Thus } \bar{\mathbf{h}}_i^{su} = \begin{bmatrix} h_k^{su} g_{1,i} \sqrt{q_k} \\ h_k^{su} g_{2,i} \sqrt{q_k} \\ \vdots \\ h_k^{su} g_{N,i} \sqrt{q_k} \end{bmatrix}. \mathbf{n}_k^{su} \sim$$

$CN(0, N_0 \mathbf{I}_N)$ is additive white Gaussian noise (AWGN) vector. “ $\odot$ ” represents the dot multiplication between matrices.

After receiving the superposed signal in downlink, the receiver uses successive interference cancellation (SIC) algorithm to distinguish and separate the signals of different users according to the corresponding power scaling factor [20]. However, the receiver cannot decode and separate user signal perfectly in the actual system due to various environmental factors or hardware defects. Imperfect SIC (ISIC) generates residual interference that affects system performance.

For the received signal-to-interference-plus-noise (SINR) of U_k , in addition to additive noise, there are two types of interference including the interference of uncoded users and the residual interference of decoded users. So according to [21], the SINR for U_k can be represented by

$$\mathfrak{R}_k = \alpha_k P (\bar{\mathbf{h}}_k^{su})^T \left[N_0 \mathbf{I}_N + \sum_{i=k+1}^K \alpha_i P \bar{\mathbf{h}}_i^{su} (\bar{\mathbf{h}}_i^{su})^H + \sum_{i=1}^{k-1} \kappa_{i,k} \alpha_i P \bar{\mathbf{h}}_i^{su} (\bar{\mathbf{h}}_i^{su})^H \right]^{-1} \bar{\mathbf{h}}_k^{su}, \quad (8)$$

$\kappa_{i,k}$ is the residual interference coefficient that represents the residual interference of U_k on U_i with ISIC.

To simplify the subsequent derivation, let $K = 3$ and $N = 2$, that is, choose three users as a PDMA cluster multiplexed on two REs with $G_{2 \times 3}^{PDMA} = \begin{bmatrix} 110 \\ 101 \end{bmatrix}$, where U_3 is an incremental relay.

As the first decoded user, U_1 decodes its own signal directly, so no residual interference exists in the SINR of U_1 . The received SINR can be obtained as

$$\mathfrak{R}_1^{DP} = \frac{\alpha_1 \bar{r} q_1 |h_1^{su}|^2}{1 + \alpha_2 \bar{r} q_1 |h_1^{su}|^2} + \frac{\alpha_1 \bar{r} q_1 |h_1^{su}|^2}{1 + \alpha_3 \bar{r} q_1 |h_1^{su}|^2}, \quad (9)$$

where $\bar{r} = \frac{P}{N_0}$.

U_2 with better channel quality first decodes the signal of U_1 using ISIC, therefore, the SINR is obtained as

$$\mathfrak{R}_{2 \rightarrow 1}^{DP} = \frac{\alpha_1 \bar{r} q_2 |h_2^{su}|^2}{1 + \alpha_2 \bar{r} q_2 |h_2^{su}|^2} + \frac{\alpha_1 \bar{r} q_2 |h_2^{su}|^2}{1 + \alpha_3 \bar{r} q_2 |h_2^{su}|^2}. \quad (10)$$

Then U_2 subtracts the decoded signal of U_1 imperfectly and then decodes its own signal, thus the received SINR including residual interference and undecoded interference is obtained as

$$\mathfrak{R}_2^{DP} = \frac{(\alpha_2 H |h_2^{su}|^2 + \alpha_3 \alpha_2 H^2 |h_2^{su}|^4 + \kappa_{1,2} \alpha_1 \alpha_2 H^2 |h_2^{su}|^4)}{1 + \alpha_3 H |h_2^{su}|^2 + 2\kappa_{1,2} \alpha_1 H |h_2^{su}|^2 + \kappa_{1,2} \alpha_1 \alpha_3 H^2 |h_2^{su}|^4}, \quad (11)$$

where $H = \bar{r} q_2$, $\bar{r} = \frac{P}{N_0}$.

U_3 with the best channel first decodes the signal of U_1 , thus the received SINR is derived as

$$\mathfrak{R}_{3 \rightarrow 1}^{DP} = \frac{\alpha_1 \bar{r} q_3 |h_3^{su}|^2}{1 + \alpha_2 \bar{r} q_3 |h_3^{su}|^2} + \frac{\alpha_1 \bar{r} q_3 |h_3^{su}|^2}{1 + \alpha_3 \bar{r} q_3 |h_3^{su}|^2}. \quad (12)$$

Then U_3 subtracts the decoded data of U_1 imperfectly and decodes the signal of U_2 , thus the received SINR is obtained as

$$\mathfrak{R}_{3 \rightarrow 2}^{DP} = \frac{\alpha_2 B |h_3^{su}|^2 + \kappa_{1,2} \alpha_1 \alpha_2 B^2 |h_3^{su}|^4}{1 + \alpha_3 B |h_3^{su}|^2 + 2\kappa_{1,2} \alpha_1 B |h_3^{su}|^2 + \kappa_{1,2} \alpha_1 \alpha_3 B^2 |h_3^{su}|^4}, \quad (13)$$

where $B = \bar{r} q_3$.

Finally, U_3 subtracts the decoded data of U_2 imperfectly and decodes its own signal, the SINR including residual interference and undecoded interference is obtained as

$$\mathfrak{R}_3^{DP} = \frac{(\alpha_3 B |h_3^{su}|^2 + \kappa_{1,3} \alpha_1 \alpha_3 B^2 |h_3^{su}|^4 + \kappa_{2,3} \alpha_2 \alpha_3 B^2 |h_3^{su}|^4)}{1 + 2\kappa_{1,3} \alpha_1 B |h_3^{su}|^2 + \kappa_{2,3} \alpha_2 B |h_3^{su}|^2 + \kappa_{2,3} \kappa_{1,3} \alpha_1 \alpha_2 B^2 |h_3^{su}|^4}. \quad (14)$$

If any user returns a NACK, the network boots into CPDMA mode. With the forwarding of U_3 , the signal transmission process of the weak users is divided into two time slots. In the first phase, the superposed signal is transmitted from the satellite to U_3 , and the signal form is the same as that in DPDMA.

In the first phase, U_3 executes PS protocol, where ρ of the received power is used for energy harvesting, and $1 - \rho$ is used for data decoding. So the SINR can be expressed as

$$\mathfrak{R}_{3 \rightarrow 1}^{CP1} = \frac{\alpha_1 (1 - \rho) \bar{r} q_3 |h_3^{su}|^2}{1 + \alpha_2 (1 - \rho) \bar{r} q_3 |h_3^{su}|^2} + \frac{\alpha_1 (1 - \rho) \bar{r} q_3 |h_3^{su}|^2}{1 + \alpha_3 (1 - \rho) \bar{r} q_3 |h_3^{su}|^2}, \quad (15)$$

$$\mathfrak{R}_{3 \rightarrow 2}^{CP1} = \frac{\alpha_2 X |h_3^{su}|^2 + \kappa_{1,2} \alpha_1 \alpha_2 X^2 |h_3^{su}|^4}{1 + \alpha_3 X |h_3^{su}|^2 + 2\kappa_{1,2} \alpha_1 X |h_3^{su}|^2 + \kappa_{1,2} \alpha_1 \alpha_3 X^2 |h_3^{su}|^4}, \quad (16)$$

$$\mathfrak{R}_3^{CP1} = \frac{\alpha_3 X |h_3^{su}|^2 + \kappa_{1,3} \alpha_1 \alpha_3 X^2 |h_3^{su}|^4 + \kappa_{2,3} \alpha_2 \alpha_3 X^2 |h_3^{su}|^4}{1 + 2\kappa_{1,3} \alpha_1 X |h_3^{su}|^2 + \kappa_{2,3} \alpha_2 X |h_3^{su}|^2 + \kappa_{2,3} \alpha_2 \kappa_{1,3} \alpha_1 X^2 |h_3^{su}|^4}, \quad (17)$$

where $X = (1 - \rho) \bar{r} q_3$.

During the second phase, U_3 forwards the re-decoded data to U_1 or U_2 with the harvested energy. So the received power at U_i ($i = 1, 2$) can be expressed as

$$P_{ri} = E_i P q_3 \eta \rho |h_3^{su}|^2. \quad (18)$$

Assuming that selective combining (SC) is adopt, the received SINR at U_i is obtained as

$$\mathfrak{R}_i^{CP2} = \frac{P_{ri} |h_i^{uu}|^2}{N_u} = \bar{r} E_i q_3 \eta \rho |h_3^{su}|^2 |h_i^{uu}|^2, \quad (19)$$

where $\bar{r} = \frac{P}{N_u}$, and N_u is average power of AWGN for the terrestrial links.

3. Performance Analysis

In the proposed PSEIR-CPDMA system, the target rate of U_i for successful decoding is R_i (bits per channel use, i.e. bpcu). In DPDMA mode, the corresponding SINR threshold is $T_i = 2^{R_i} - 1$. The threshold is set to be $T'_i = 2^{2R_i} - 1$ in CPDMA, since data transmission requires two time slots.

3.1 Outage Probability

3.1.1 The Exact OP of PSEIR-CPDMA

In the system, data is transmitted in DPDMA or CPDMA mode. For U_1 and U_2 , the outage occurs when neither the direct link nor the forwarding link is valid. As an incremental relay with only direct link, U_3 is interrupted when it fails to decode data in both mode. The OP of U_i is expressed as

$$O_i = 1 - (P_i^{DP} + P_i^{CP}), \quad (20)$$

where P_i^{DP} and P_i^{CP} are the probabilities of successful decoding of U_i in DPDMA and CPDMA modes respectively.

$$\begin{aligned} P_1^{DP} &= \Pr\{\mathfrak{R}_1^{DP} > T_1\} \\ &= \Pr\left\{[(\alpha_3 + \alpha_2) \alpha_1 - \alpha_3 \alpha_2 T_1] (\bar{r} q_1)^2 |h_1^{su}|^4 \right. \\ &\quad \left. + [2\alpha_1 - (\alpha_2 + \alpha_3) T_1] (\bar{r} q_1) |h_1^{su}|^2 - T_1 > 0\right\} \\ &= \begin{cases} F_{|h_1^{su}|^2}(\hat{x}_2) - F_{|h_1^{su}|^2}(\hat{x}_1) & a < 0 \\ 1 + F_{|h_1^{su}|^2}(\hat{x}_1) - F_{|h_1^{su}|^2}(\hat{x}_2) & a > 0 \end{cases}, \\ &= \begin{cases} 0 & a < 0 \\ 1 - \rho_1 \sum_{k=0}^{\tau_1-1} \Pi_{1,k} \gamma(k+1, (\beta_1 - \delta_1) \hat{x}_2) & a > 0 \end{cases} \end{aligned} \quad (21)$$

in which $a = (\alpha_3 + \alpha_2) \alpha_1 - \alpha_3 \alpha_2 T_1$, $b = 2\alpha_1 - (\alpha_2 + \alpha_3) T_1$, $x_1 = \frac{-b - \sqrt{b^2 + 4T_1 a}}{2a \bar{r} q_1}$, $x_2 = \frac{-b + \sqrt{b^2 + 4T_1 a}}{2a \bar{r} q_1} = \Upsilon \bar{r}^{-1}$, $\hat{x}_1 = \max\{0, x_1\} = 0$, $\hat{x}_2 = \max\{0, x_2\} = \begin{cases} 0 & a < 0 \\ x_2 & a > 0 \end{cases}$.

$$\begin{aligned} P_1^{CP} &= \Pr\{\mathfrak{R}_1^{DP} < T_1, \mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1, \mathfrak{R}_1^{CP2} > T'_1\} \\ &= (1 - \Pr\{\mathfrak{R}_1^{DP} > T_1\}) \Pr\{\mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1\} (1 - \Pr\{\mathfrak{R}_1^{CP2} < T'_1\}), \end{aligned} \quad (22)$$

where

$$\Pr\{\mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1\} = \begin{cases} 0 & \bar{a} < 0 \\ 1 - \rho_3 \sum_{k=0}^{\tau_3-1} \Pi_{3,\gamma}(k+1, (\beta_3 - \delta_3) \bar{x}_2) & \bar{a} > 0 \end{cases}, \quad (23)$$

in which $\bar{a} = (\alpha_3 + \alpha_2)\alpha_1 - \alpha_3\alpha_2T'_1$, $\bar{b} = 2\alpha_1 - (\alpha_2 + \alpha_3)T'_1$,

$$\bar{x}_2 = \frac{-\bar{b} + \sqrt{\bar{b}^2 + 4T'_1\bar{a}}}{2\bar{a}q_3(1-\rho)} = \bar{\Upsilon}\bar{r}^{-1}.$$

According to [16], express $\gamma(1+n, x) = n! \left[1 - e^{-x} \left(\sum_{m=0}^n \frac{x^m}{m!} \right) \right]$. It follows that

$$\begin{aligned} \Pr\{\mathfrak{R}_1^{CP2} < T'_1\} &= \Pr\left\{|h_3^{su}|^2 < \frac{T'_1}{\tilde{r}E_1q_3\eta\rho|h_1^{uu}|^2}\right\} \\ &= \Lambda_1 \sum_{k=0}^{\tau_3-1} \Theta_{3,k} \left[\frac{1}{\varepsilon_1^{m_1}} \Gamma(m_1) - 2 \sum_{m=0}^k \frac{[(\beta_3 - \delta_3)\Psi_1]^{\frac{m_1+m}{2}}}{m! \varepsilon_1^{\frac{m_1-m}{2}}} \right. \\ &\quad \left. K_{m_1-m}(2\sqrt{(\beta_3 - \delta_3)\Psi_1\varepsilon_1}) \right], \quad (24) \end{aligned}$$

where $\Psi_1 = \frac{T'_1}{\tilde{r}E_1q_3\eta\rho}$, $\Lambda_1 = \frac{\rho_3\varepsilon_1^{m_1}}{\Gamma(m_1)}$, $\Theta_{i,k} = k!\Pi_{i,k}$, $K_v(z) = \int_0^\infty e^{-z \cosh t} \cosh vt dt$ is Bessel functions of imaginary argument.

$$P_2^{DP} = \Pr\{\mathfrak{R}_2^{DP} > T_2\} = \begin{cases} 0 & \hat{a} < 0 \\ 1 - F_{|h_2^{su}|^2}(\hat{x}_2) & \hat{a} > 0 \end{cases}, \quad (25)$$

in which $\hat{a} = \alpha_3\alpha_2 + \kappa_{1,2}\alpha_1\alpha_2 - \kappa_{1,2}\alpha_1\alpha_3T_2$, $\hat{b} = \alpha_2 - \alpha_3T_2 - 2\kappa_{1,2}\alpha_1T_2$, $\hat{x}_2 = \frac{-\hat{b} + \sqrt{\hat{b}^2 + 4T_2\hat{a}}}{2\hat{a}H} = \hat{\Upsilon}\hat{r}^{-1}$.

$$\begin{aligned} P_2^{CP} &= \Pr\{\mathfrak{R}_2^{DP} < T_2, \mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1, \mathfrak{R}_{3 \rightarrow 2}^{CP1} > T'_2, \mathfrak{R}_2^{CP2} > T'_2\} \\ &= (1 - \Pr\{\mathfrak{R}_2^{DP} > T_2\}) \Pr\{\mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1\} \Pr\{\mathfrak{R}_{3 \rightarrow 2}^{CP1} > T'_2\} \\ &\quad (1 - \Pr\{\mathfrak{R}_2^{CP2} > T'_2\}), \quad (26) \end{aligned}$$

the expressions of each part in (26) are as follows respectively.

$$\Pr\{\mathfrak{R}_{3 \rightarrow 2}^{CP1} > T'_2\} = \begin{cases} 0 & \check{a} \leq 0 \\ 1 - F_{|h_3^{su}|^2}(\check{x}_2) & \check{a} \geq 0 \end{cases}, \quad (27)$$

in which $\check{a} = \kappa_{1,2}\alpha_1\alpha_2 - \kappa_{1,2}\alpha_1\alpha_3T'_2$, $\check{b} = \alpha_2 - \alpha_3T'_2 - 2\kappa_{1,2}\alpha_1T'_2$, $\check{x}_2 = \frac{-\check{b} + \sqrt{\check{b}^2 + 4T'_2\check{a}}}{2\check{a}X} = \check{\Upsilon}\check{r}^{-1}$.

$$\begin{aligned} \Pr\{\mathfrak{R}_2^{CP2} < T'_2\} &= \Pr\left\{\tilde{r}E_2q_3\eta\rho|h_3^{su}|^2|h_2^{uu}|^2 < T'_2\right\} \\ &= \Lambda_2 \sum_{k=0}^{\tau_2-1} \Theta_{2,k} \left[\frac{1}{\varepsilon_2^{m_2}} \Gamma(m_2) - 2 \sum_{m=0}^k \frac{[(\beta_3 - \delta_3)\Psi_2]^{\frac{m_2+m}{2}}}{m! (\varepsilon_2)^{\frac{m_2-m}{2}}} \right. \\ &\quad \left. K_{m_2-m}(2\sqrt{(\beta_3 - \delta_3)\Psi_2\varepsilon_2}) \right], \quad (28) \end{aligned}$$

where $\Psi_2 = \frac{T'_2}{\tilde{r}E_2q_3\eta\rho}$, $\Lambda_2 = \frac{\rho_3\varepsilon_2^{m_2}}{\Gamma(m_2)}$.

$$\begin{aligned} P_3^{DP} &= \Pr\{\mathfrak{R}_1^{DP} > T_1, \mathfrak{R}_2^{DP} > T_2, \mathfrak{R}_{3 \rightarrow 1}^{DP} > T_1, \mathfrak{R}_{3 \rightarrow 2}^{DP} > T_2, \mathfrak{R}_3^{DP} > T_3\} \\ &= \Pr\{\mathfrak{R}_1^{DP} > T_1\} \Pr\{\mathfrak{R}_2^{DP} > T_2\} \Pr\{\mathfrak{R}_{3 \rightarrow 1}^{DP} > T_1\} \Pr\{\mathfrak{R}_{3 \rightarrow 2}^{DP} > T_2\} \\ &\quad \Pr\{\mathfrak{R}_3^{DP} > T_3\}. \quad (29) \end{aligned}$$

$$\Pr\{\mathfrak{R}_{3 \rightarrow 1}^{DP} > T_1\} = \begin{cases} 0 & a < 0 \\ 1 - F_{|h_3^{su}|^2}(\tilde{x}_2) & a > 0 \end{cases}, \quad (30)$$

in which $\tilde{x}_2 = \frac{-b + \sqrt{b^2 + 4T_1a}}{2\tilde{a}q_3} = \tilde{\Upsilon}\tilde{r}^{-1}$.

$$\Pr\{\mathfrak{R}_{3 \rightarrow 2}^{DP} > T_2\} = \begin{cases} 0 & \vec{a} < 0 \\ 1 - F_{|h_3^{su}|^2}(\vec{x}_2) & \vec{a} > 0 \end{cases}, \quad (31)$$

in which $\vec{a} = \kappa_{1,2}\alpha_1\alpha_2 - \kappa_{1,2}\alpha_1\alpha_3T_2$, $\vec{b} = \alpha_2 - \alpha_3T_2 - 2\kappa_{1,2}\alpha_1T_2$, $\vec{x}_2 = \frac{-\vec{b} + \sqrt{\vec{b}^2 + 4T_2\vec{a}}}{2\vec{a}B} = \vec{\Upsilon}\vec{r}^{-1}$.

$$\Pr\{\mathfrak{R}_3^{DP} > T_3\} = \begin{cases} 0 & \overleftarrow{a} < 0 \\ 1 - F_{|h_3^{su}|^2}(\overleftarrow{x}_2) & \overleftarrow{a} > 0 \end{cases}, \quad (32)$$

in which $\overleftarrow{a} = \kappa_{1,3}\alpha_1\alpha_3 + \kappa_{2,3}\alpha_2\alpha_3 - \kappa_{2,3}\kappa_{1,3}\alpha_1\alpha_2T_3$, $\overleftarrow{b} = \alpha_3 - 2\kappa_{1,3}\alpha_1T_3 - \kappa_{2,3}\alpha_2T_3$, $\overleftarrow{x}_2 = \frac{-\overleftarrow{b} + \sqrt{\overleftarrow{b}^2 + 4T_3\overleftarrow{a}}}{2\overleftarrow{a}B} = \overleftarrow{\Upsilon}\overleftarrow{r}^{-1}$.

$$\begin{aligned} P_3^{CP} &= \Pr\{\mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1, \mathfrak{R}_{3 \rightarrow 2}^{CP1} > T'_2, \mathfrak{R}_3^{CP1} > T'_3\} \\ &\quad (1 - \Pr\{\mathfrak{R}_1^{DP} > T_1, \mathfrak{R}_2^{DP} > T_2\}) \\ &= \Pr\{\mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1\} \Pr\{\mathfrak{R}_{3 \rightarrow 2}^{CP1} > T'_2\} \Pr\{\mathfrak{R}_3^{CP1} > T'_3\} \\ &\quad [1 - \Pr\{\mathfrak{R}_1^{DP} > T_1\} \Pr\{\mathfrak{R}_2^{DP} > T_2\}]. \quad (33) \end{aligned}$$

$$\Pr\{\mathfrak{R}_3^{CP1} > T'_3\} = \begin{cases} 0 & \overleftrightarrow{a} < 0 \\ 1 - F_{|h_3^{su}|^2}(\overleftrightarrow{x}_2) & \overleftrightarrow{a} > 0 \end{cases}, \quad (34)$$

in which $\overleftrightarrow{a} = \kappa_{1,3}\alpha_1\alpha_3 + \kappa_{2,3}\alpha_2\alpha_3 - \kappa_{2,3}\alpha_2\kappa_{1,3}\alpha_1T'_3$, $\overleftrightarrow{b} = \alpha_3 - 2\kappa_{1,3}\alpha_1T'_3 - \kappa_{2,3}\alpha_2T'_3$, $\overleftrightarrow{x}_2 = \frac{-\overleftrightarrow{b} + \sqrt{\overleftrightarrow{b}^2 + 4T'_3\overleftrightarrow{a}}}{2\overleftrightarrow{a}X} = \overleftrightarrow{\Upsilon}\overleftrightarrow{r}^{-1}$.

Thus, the average outage probability of the system can be expressed as

$$O_{av} = \frac{1}{3} (O_1 + O_2 + O_3). \quad (35)$$

3.1.2 The Asymptotic OP of PSEIR-CPDMA

In order to make a comprehensive analysis of the performance of PDMA users, we analyze the asymptotic OP of each user at the high SINR, and then obtain the diversity order G_i^d and the coding gain G_i^c [22].

For the satellite-terrestrial link, expand $\gamma(k+1, \beta_i u)$ in series [16], and we can get

$$\gamma(k+1, \beta_i u) = \sum_{k=0}^{\infty} \frac{(-1)^n (\beta_i u)^{k+1+n}}{n!(k+1+n)} \approx \frac{(\beta_i u)^{k+1}}{k+1} \Big|_{u \rightarrow 0}. \quad (36)$$

Substitute (36) into (3), so the asymptotic CDF of $|h_{n,i}^{su}|^2$ can be expressed as

$$F_{|h_{n,i}^{su}|^2}(u) \approx \rho_i \sum_{k=0}^{\infty} \frac{(\tau_i)_k \delta_i^k}{(k!)^2 \beta_i^{k+1}} \frac{(\beta_i u)^{k+1}}{k+1} \approx \rho_i u. \quad (37)$$

Based on (37), as $\bar{r} \rightarrow \infty$, $\tilde{r} \rightarrow \infty$ and $\tilde{r} = \vartheta \bar{r}$, we can get the asymptotic expressions of each probability as follows:

$$\Pr^{\infty} \{ \mathfrak{R}_1^{DP} > T_1 \} \approx \begin{cases} 0 & a < 0 \\ 1 - \rho_1 \hat{x}_2 & a > 0 \end{cases}, \quad (38)$$

$$\Pr^{\infty} \{ \mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1 \} \approx \begin{cases} 0 & \bar{a} < 0 \\ 1 - \rho_3 \bar{x}_2 & \bar{a} > 0 \end{cases}, \quad (39)$$

$$\begin{aligned} \Pr^{\infty} \{ \mathfrak{R}_1^{CP2} < T'_1 \} &= \int_0^{\infty} F_{|h_{n,i}^{su}|^2} \left(\frac{\Psi_1}{x} \right) f_{|h_{n,i}^{su}|^2}(x) dx \\ &\approx \int_0^{\infty} \rho_3 \frac{\Psi_1}{x} f_{|h_{n,i}^{su}|^2}(x) dx = \frac{\rho_3 \Psi_1 \varepsilon_1^{m_1}}{\Gamma(m_1)} \frac{\Gamma(m_1 - 1)}{\varepsilon_1^{m_1 - 1}} = \frac{\rho_3 \Psi_1 \varepsilon_1}{m_1 - 1}. \end{aligned} \quad (40)$$

The asymptotic OP of U_1 is

$$\begin{aligned} O_1^{\infty} &= 1 - (P_1^{DP} + P_1^{CP}) \\ &\approx \begin{cases} \left(\rho_1 \rho_3 \tilde{\Upsilon} \bar{\Upsilon} + \frac{\rho_1 \rho_3 \tilde{\Upsilon} T'_1 \varepsilon_1}{E_1 q_3 \eta \rho \vartheta (m_1 - 1)} \right) \bar{r}^{-2} & \text{if } (a > 0 \& \bar{a} > 0) \\ \rho_1 \tilde{\Upsilon} \bar{r}^{-1} & \text{elseif } (a > 0 \& \bar{a} < 0) \\ 1 & \text{else} \end{cases}. \end{aligned} \quad (41)$$

So the diversity order G_1^d and the coding gain G_1^c are $G_1^d =$

$$\begin{cases} 2 & \text{if } (a > 0 \& \bar{a} > 0) \\ 1 & \text{elseif } (a > 0 \& \bar{a} < 0) \text{ and} \\ 0 & \text{else} \end{cases}$$

$$G_1^c = \begin{cases} \left(\rho_1 \rho_3 \tilde{\Upsilon} \bar{\Upsilon} + \frac{\rho_1 \rho_3 \tilde{\Upsilon} T'_1 \varepsilon_1}{E_1 q_3 \eta \rho \vartheta (m_1 - 1)} \right) & \text{if } (a > 0 \& \bar{a} > 0) \\ \rho_1 \tilde{\Upsilon} & \text{elseif } (a > 0 \& \bar{a} < 0) \\ 1 & \text{else} \end{cases}.$$

The asymptotic OP of U_2 is

$$\begin{aligned} O_2^{\infty} &= 1 - (P_2^{DP} + P_2^{CP}) \\ &\approx \begin{cases} \left(\rho_2 \rho_3 \tilde{\Upsilon} \bar{\Upsilon} + \rho_2 \rho_3 \tilde{\Upsilon} \bar{\Upsilon} + \frac{\rho_2 \rho_3 \tilde{\Upsilon} T'_2 \varepsilon_2}{E_2 q_3 \eta \rho \vartheta (m_2 - 1)} \right) \bar{r}^{-2} & \text{if } (\hat{a} > 0 \& \bar{a} > 0 \& \check{a} > 0) \\ \rho_2 \tilde{\Upsilon} \bar{r}^{-1} & \text{elseif } (\hat{a} > 0 \& \check{a} < 0) \\ 1 & \text{else} \end{cases}. \end{aligned} \quad (42)$$

So the diversity order G_2^d and the coding gain G_2^c

$$\text{are } G_2^d = \begin{cases} 2 & \text{if } (\hat{a} > 0 \& \bar{a} > 0 \& \check{a} > 0) \\ 1 & \text{elseif } (\hat{a} > 0 \& \check{a} < 0) \\ 0 & \text{else} \end{cases} \quad \text{and } G_2^c =$$

$$\begin{cases} \rho_2 \rho_3 \tilde{\Upsilon} \bar{\Upsilon} + \rho_2 \rho_3 \tilde{\Upsilon} \bar{\Upsilon} + \frac{\rho_2 \rho_3 \tilde{\Upsilon} T'_2 \varepsilon_2}{E_2 q_3 \eta \rho \vartheta (m_2 - 1)} & \text{if } (\hat{a} > 0 \& \bar{a} > 0 \& \check{a} > 0) \\ \rho_2 \tilde{\Upsilon} & \text{elseif } (\hat{a} > 0 \& \check{a} < 0) \\ 1 & \text{else} \end{cases}.$$

The asymptotic OP of U_3 is derived as follows:

$$P_3^{DP} \approx \begin{cases} 1 - \rho_1 \hat{x}_2 - \rho_2 \bar{x}_2 - \rho_3 \check{x}_2 - \rho_3 \bar{x}_2 & \text{if } M \\ 0 & \text{else} \end{cases}, \quad (43)$$

in which M : $(a > 0 \& \bar{a} > 0 \& \check{a} > 0 \& \bar{a} > 0)$.

$$P_3^{CP} \approx \begin{cases} (1 - \rho_3 \bar{x}_2)(1 - \rho_3 \check{x}_2)(1 - \rho_3 \bar{x}_2)[1 - (1 - \rho_1 \hat{x}_2)(1 - \rho_2 \bar{x}_2)] & \text{if } M' \\ 0 & \text{else} \end{cases} \quad (44)$$

in which M' : $(\bar{a} > 0 \& \check{a} > 0 \& \bar{a} > 0)$.

$$\begin{aligned} O_3^{\infty} &= 1 - (P_3^{DP} + P_3^{CP}) \\ &= \begin{cases} \rho_3 (\tilde{\Upsilon} + \bar{\Upsilon} + \bar{\Upsilon}) \bar{r}^{-1} & \text{if } S \\ \left(\rho_1 \tilde{\Upsilon} + \rho_2 \hat{\Upsilon} + \rho_3 \tilde{\Upsilon} + \rho_3 \bar{\Upsilon} + \rho_3 \bar{\Upsilon} \right) \bar{r}^{-1} & \text{elseif } S' \\ 1 & \text{else} \end{cases}, \end{aligned} \quad (45)$$

in which S : $(a > 0 \& \bar{a} > 0 \& \check{a} > 0 \& \bar{a} > 0 \& \check{a} > 0 \& \bar{a} > 0 \& \check{a} > 0)$, S' : $(a > 0 \& \bar{a} > 0 \& \check{a} > 0 \& \bar{a} > 0 \& \check{a} > 0 \& \bar{a} < 0 \parallel \check{a} < 0 \parallel \bar{a} < 0)$.

So the diversity order G_3^d and the coding gain G_3^c are

$$\begin{aligned} G_3^d &= \begin{cases} 1 & \text{if } M \\ 0 & \text{else} \end{cases} \text{ and} \\ G_3^c &= \begin{cases} \rho_3 (\tilde{\Upsilon} + \bar{\Upsilon} + \bar{\Upsilon}) & \text{if } S \\ \left(\rho_1 \tilde{\Upsilon} + \rho_2 \hat{\Upsilon} + \rho_3 \tilde{\Upsilon} + \rho_3 \bar{\Upsilon} + \rho_3 \bar{\Upsilon} \right) & \text{elseif } S' \\ 1 & \text{else} \end{cases}. \end{aligned}$$

The diversity gain of U_3 is less than or equal to U_2 and U_1 , whose values are related to the power scaling factor α_i and the target rate R_i . The greater R_i is, the smaller the diversity gain and the coding gain are.

3.1.3 OP of PS-CPDMA

In conventional cooperative PDMA with PS (PS-CPDMA), U_3 is always in cooperative mode without knowing the link status of U_1 and U_2 . And U_1 and U_2 selectively combine signals from the satellite and U_3 in the second phase. The OPs of each user can be expressed as

$$O_{c1} = \Pr \{ \max \{ \mathfrak{R}_1^{CP2}, \mathfrak{R}_1^{DP} \} < T'_1 \}, \quad (46)$$

$$O_{c2} = \Pr \{ \max \{ \mathfrak{R}_2^{CP2}, \mathfrak{R}_2^{DP} \} < T'_2 \}, \quad (47)$$

$$\begin{aligned} O_{c3} &= \Pr \{ \mathfrak{R}_{3 \rightarrow 1}^{CP1} < T'_1 \} + \Pr \{ \mathfrak{R}_{3 \rightarrow 2}^{CP1} < T'_2 / \mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1 \} \\ &\quad + \Pr \{ \mathfrak{R}_3^{CP1} < T'_3 / \mathfrak{R}_{3 \rightarrow 2}^{CP1} > T'_2, \mathfrak{R}_{3 \rightarrow 1}^{CP1} > T'_1 \} \end{aligned} \quad (48)$$

3.2 Throughput

3.2.1 Throughput of PSEIR-CPDMA

In the delay-limited transmission, the users are served at a constant rate. In DPDMA mode, the successful rates are R_1 and R_2 for U_1 and U_2 . However, the corresponding rates are $\frac{R_1}{2}$ and $\frac{R_2}{2}$ respectively in CPDMA. Therefore, the total throughput of the system can be expressed as

$$T^{IR} = \sum_{i=0}^3 P_i^{DP} R_i + \sum_{i=0}^3 P_i^{CP} R_i/2. \quad (49)$$

3.2.2 Throughput of PS-CPDMA

In PS-CPDMA, the users always decode data in CPDMA mode. So the sum throughput of PS-CPDMA system is expressed as

$$T^{CR} = \sum_{i=0}^3 (1 - O_{ci}) R_i/2. \quad (50)$$

4. Numerical Results

To verify the performance of the proposed scheme, two communication scenarios of PSEIR-CPDMA and PS-CPDMA with three users are simulated using MATLAB. In this section, not only the correctness and the feasibility of the proposed PSEIR-CPDMA scheme are verified from the perspectives of outage probability and system throughput, but also the energy efficiencies of the two scenarios are compared. The three users are supposed to experience heavy shadowing (HS), average shadowing (AS), and light shadowing (LS) with parameters shown in Table 1.

When $R = 0.8$ and the residual interference coefficients $\kappa_{1,2} = \kappa_{1,3} = \kappa_{2,3} = 0.1$, the comparison of outage performance with perfect SIC and imperfect SIC schemes for the three users is shown in Fig. 2. At the R value, all the three users work in the CPDMA mode. As an IR, $\rho\%$ power of U_3 needs to be used for data forwarding, meanwhile, U_3 is assigned the minimum power scaling factor according to DTPA because of its optimal channel quality, which results in its worst OP performance. However, due to the better channel quality and the timely forwarding by U_3 , U_2 achieves optimal outage performance. As seen from the figure, the outage probability with ISIC is higher than that of SIC because the SINRs value of U_2 and U_3 decrease due to the presence of residual interference. U_1 can be decoded first with no residual interference, so its outage performance remains unchanged in both SIC and ISIC. It can be observed that the asymptotic OP excellently coincides with the analytical OP in the high SINR range.

Figure 3 shows the relationship between the outage probability and the target rate R for three users in PSEIR-CPDMA. As can be seen from the figure, U_3 as a relay node

Table 1 PSEIR-CPDMA performance simulation assumptions.

Attributes	Values of assumptions
Carrier frequency f	2GHz
Satellite antenna gain G_s	54dBi
(τ_i, b_i, Ω_i) in HS, AS, LS	(2, 0.063, 0.0005), (5, 0.251, 0.279), (10, 0.126, 0.835)
$S \rightarrow U_j$ angle with 3dB φ_{j3dB}	0.3°
$S \rightarrow U_j$ angle φ_j	0.2°

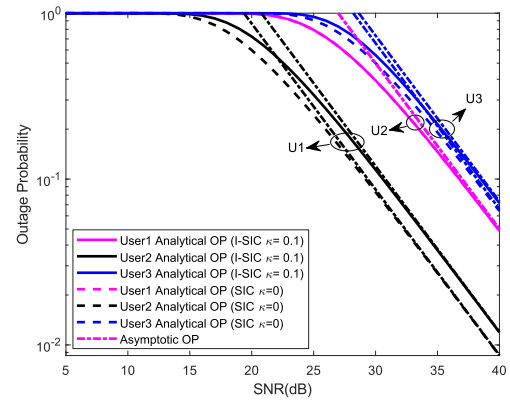


Fig. 2 Outage probability vs SNR ($\rho = 0.2$, $\eta = 0.7$).

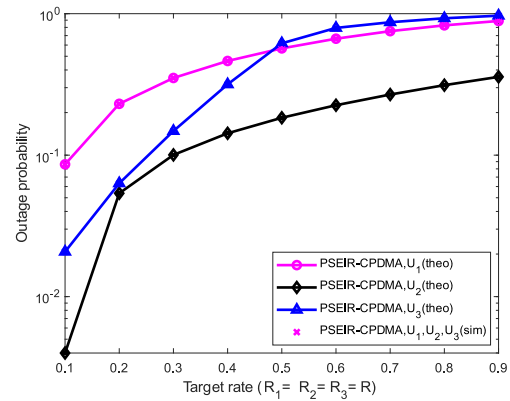


Fig. 3 Outage probability vs R ($\rho = 0.2$, $\eta = 0.7$, $\bar{r} = 30$ dB).

has the better outage performance with the lower R even though it is assigned the lowest power scaling factor. However, with the value of R increasing, the three users enter the cooperation mode with higher and higher SINR threshold, so the outage performance decreases. The relay user uses the harvested energy to forward data for the other two users in CPDMA that makes the outage performance deterioration rate of U_3 higher than the other two users. Thus the outage performance of U_3 finally becomes the worst due to the lowest power scaling factor and power splitting.

As shown in Fig. 4, the outage performance in PSEIR-CPDMA is apparently improved compared with the conven-

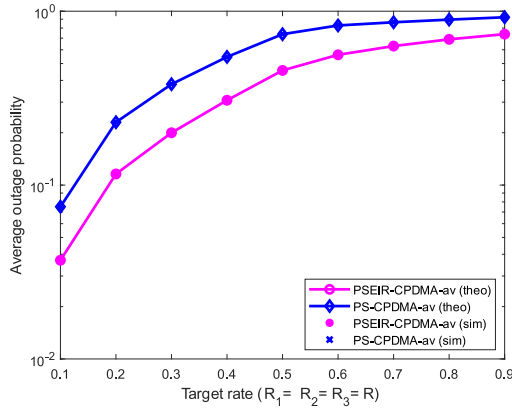


Fig. 4 Average outage probability vs R ($\rho = 0.2$, $\eta = 0.7$, $\bar{r} = 30$ dB).

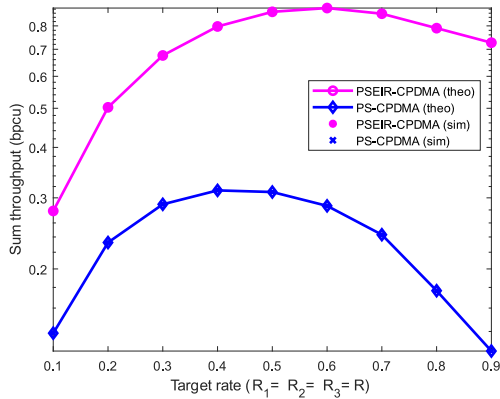


Fig. 5 Sum throughput vs R ($\rho = 0.2$, $\eta = 0.7$, $\bar{r} = 30$ dB).

tional PS-CPDMA under the same condition. This is because that PSEIR-CPDMA uses both DPDMA and CPDMA for data transmission, while PS-CPDMA only adopts CPDMA.

Figure 5 shows the result of the sum throughput versus target rate R for PSEIR-CPDMA. The sum throughput first increases and then decreases with the value of R . When R increases, the probability P_i^{DP} and P_i^{CP} decreases with the SINR threshold increasing. Thus an optimal R can be seen to maximize the system throughput. Furthermore, the result also shows that the throughput performance of PSEIR-CPDMA is significantly better than that of PS-CPDMA.

The change of sum throughput against ρ is shown in Fig. 6. When $\rho = 0$, the power used to forward data becomes zero, and the system only has direct link transmission. For the higher ρ , the power used for decoding data at U_3 decreases, while the power available for forwarding data increases, i.e., P_3^{CP} is lower, but P_1^{CP} and P_2^{CP} is higher. As can be seen, an optimal ρ exists related to R_i . When R_3 increases from 0.2 to 0.3 with fixed $R_1 = R_2 = 0.3$, the optimal ρ decreases, because more power is required for successful decoding at U_3 with higher SINR threshold. However, when R_i increases to a certain extent, the probability of successful decoding decreases, and the throughput is mainly affected by direct transmission which causes the optimal ρ decreases.

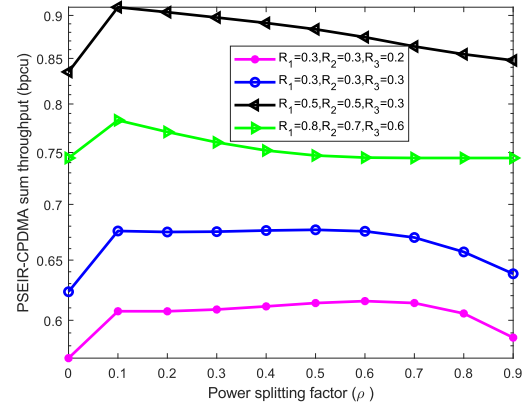


Fig. 6 Sum throughput vs power splitting factor ρ for PSEIR-CPDMA.

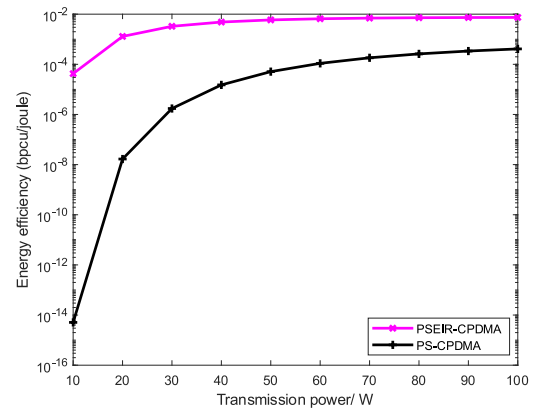


Fig. 7 Energy efficiency comparison for PSEIR-CPDMA and PS-CPDMA.

Figure 7 depicts a comparison of the energy efficiency performance of the two schemes. Energy efficiency is defined as the ratio of sum throughput to energy. Assume that the AWGN power is set as $N_0 = 0.1$ W. As can be seen from the figure, PSEIR-CPDMA has better energy efficiency than PS-CPDMA.

5. Conclusion

In the proposed three-user downlink STMN based on PDMA, the near-end user is configured as an incremental relay considering the imbalance of users' channel quality. IR converts in DPMA or CPDMA modes by judging the limited feedback. The performance of outage and throughput is investigated using ISIC detection, and its relationship with target rate and power splitting factor is analyzed. The users' diversity gains and coding gains are obtained by deducing the asymptotic OP at high SINR. Theoretical analysis and Monte-Carlo simulation show that the proposed PSEIR-CPDMA scheme can provide better system performance than the conventional PS-CPDMA.

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